# Umurima w'imboga: kuva ku butaka bwambaye ubusa kugeza mu gikoni

Uko watangiza umurima w'imboga ugaburira igikoni gikuru (galley) kandi ukagurisha ibisagutse byawo: ibihembwe, uturima, amazi, ibihingwa n'ukwezi kwa mbere, intambwe ku yindi.

> ***Ubusobanuro:*** *iyi nyandiko yahinduwe na mudasobwa (machine draft). Igomba gusuzumwa n'umuntu uvuga Ikinyarwanda kavukire mbere yo kuyiha abakozi. Iyo hari itandukaniro, kurikiza iy'Icyongereza.*

Status: amabwiriza yo ku wa 2026-10-28. Umurima, ubutaka bwawo n'amazi yawo ntibirapimwa; ingano zikurikiza igishushanyo mbonera cy'agateganyo (provisional master plan).

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## Umurima mu ncamake

| Iki | Ingano |
|---|---|
| Igice cy'imboga (vegetable zone) | 3456 m², iruhande rw'igikoni gikuru (igishushanyo mbonera) |
| Uturima (beds) | ubugari bwa m 1.2, tuzamuye cm 20, uburebure bwa m 10: hafi uturima 200, hafi 2,440 m² z'ubuso buhingwa [E] |
| Inzira (paths) | m 0.5 hagati y'uturima, m 1 hagati y'amatsinda y'uturima |
| Amazi | amatiyo y'ibitonyanga (drip lines), 2 kuri buri karima, ahabwa amazi n'ikigega cy'amazi (tank); 10 kugeza 13 m³ ku munsi mu mezi y'izuba ryinshi [E] |
| Ibyo igikoni gikuru gikenera | hafi kg 40 kugeza 50 z'imboga mu cyumweru, iyo abashyitsi bangana n'abateganyijwe mu igenamigambi fatizo (base-case occupancy) [A]; ibisigaye bigurishwa |

## Ibihembwe by'ihinga

| Igihembwe | Amezi | Ibyo umurima ukora |
|---|---|---|
| A | Nzeri kugeza Gashyantare | Kurera ingemwe (seedlings) mu gihe cy'ibyumweru 3 kugeza 8, kuzimura (transplant) igihe imvura itangiye, kuhira hakoreshejwe ibitonyanga igihe imvura ihagaze muri Mutarama na Gashyantare |
| B | Werurwe kugeza Kamena | Kwimura ingemwe muri Gashyantare no muri Werurwe, uturima tuzamuye kubera imvura ya Mata, kongera kuhira hakoreshejwe ibitonyanga guhera mu mpera za Gicurasi |
| C | Nyakanga kugeza Nzeri | Kuhira hakoreshejwe ibitonyanga gusa: buri munsi hayoyoka hafi mm 4 kugeza 5 z'amazi; gutera gusa ibyo amazi yemejwe n'igerageza ashobora kuhira |

Ibihembwe [V LOC-53]; imvura n'amazi ayoyoka i Gashora [E LOC-34, LOC-54]; igihe ingemwe zimara mu buhumbikiro [V CRP-68].

## Ukwezi kwa mbere, intambwe ku yindi (Ugushyingo)

1. **Pima ubutaka.** Fata ingero z'ubutaka (samples) ahantu 10 hatandukanye mu gice cyose, uzivange, maze uzohereze muri laboratwari y'ubutaka ya RAB: pH (ubusharire), organic carbon, azote (nitrogen), fosifore (phosphorus), imiterere y'ubutaka (texture) [V CRP-72].
2. **Pima amazi.** Pima amazi borehole (iriba ricukuwe) itanga mu gihe cy'amasaha 6, kandi usuzumishe urugero rw'amazi; umurima ukenera hafi 2 $m^3$ ku isaha mu gihe cy'amasaha 6 mu mezi y'izuba [E].
3. **Zitira igice cy'imboga** kugira ngo inyamaswa zitinjiramo, kandi usige irembo rimwe ku nzira y'ubusitani (garden walk) ijya ku gikoni gikuru.
4. **Shyira ibimenyetso aho uturima tuzaba:** uturima twa m 1.2, inzira za m 0.5, inzira za m 1 hagati y'amatsinda y'uturima 10; uturima dutambikwa ku buhaname bw'ubutaka (across the slope).
5. **Kora uturima 40 twa mbere:** cukura, uvangemo ifumbire y'imborera (compost) cyangwa iy'amatungo (manure), hafi kg 250 kuri 100 $m^2$ ku nyanya (tomatoes) [V CRP-68], hanyuma utuzamure cm 20.
6. **Shyiraho amatiyo y'ibitonyanga:** itiyo nkuru (header pipe) iva ku kigega cy'amazi, akayunguruzo (filter), n'amatiyo abiri y'ibitonyanga kuri buri karima; fungura amazi mu gihe cy'isaha imwe, ugenzure buri kenge gatonyanga (emitter).
7. **Tangiza ubuhumbikiro (nursery):** udusanduku tw'ingemwe (trays) cyangwa akarima k'ingemwe (seedbed) gatwikiriwe n'urushundura rw'igicucu (shade net), ku ngemwe za sukuma (kale), ishu (cabbage), igitunguru (onion), inyanya na puwavuro (sweet pepper).

8. **Biba imbuto mu turima, nta guhumbika:** dodo (amaranth) n'imiteja (French beans), bitanga umusaruro mu byumweru 4 kugeza 8.
9. **Shyiramo ishwagara (lime) gusa iyo ibisubizo by'igerageza bibisabye:** imboga zikenera pH 6.0 kugeza 7.0; birashoboka ko ubutaka bwo hano buri hafi ya 5.5 kugeza 6.1 [V CRP-68, E LOC-37].

## Ibihingwa

| Igihingwa | Intera yo gutera (cm) | Isarura rya mbere | Umusaruro | Ibyo kwitondera |
|---|---|---|---|---|
| Inyanya (tomato) | 100 × 50 | iminsi 60 kugeza 90 nyuma yo kwimura ingemwe | hafi 8 t/ha ku rwego rw'igihugu [V CRP-17] | Tuta absoluta (ikinyugunyugu gito, inyo zacyo zicukura amababi n'inyanya), late blight (indwara yumisha amababi n'inyanya vuba), bacterial wilt (indwara ya bagiteri ituma igihingwa kiraba) |
| Igitunguru (onion) | 20 × 10 | iminsi 90 kugeza 150 | kugeza kuri 60 t/ha (uko abacuruza ubwo bwoko babivuga) | thrips (udusimba duto cyane twonona amababi), downy mildew (uruhumbu munsi y'amababi) |
| Ishu (cabbage) | 60 × 50 kugeza 60 | iminsi 60 kugeza 105 | 25 kugeza 30 t/ha | diamondback moth (ikinyugunyugu gito, inyo zacyo zirya amababi), black rot (indwara ya bagiteri yirabuza amababi) |
| Karoti (carrot) | imirongo 20, hanyuma ugakuramo izirenze hagasigara 5 kugeza 8 | iminsi 75 kugeza 110 nyuma yo kubiba | 20 kugeza 30 t/ha | carrot fly (isazi, inyo zayo zicukura karoti), leaf blight (indwara itera amabara yumisha amababi) |
| Puwavuro (sweet pepper) | 60 × 40 kugeza 45 | iminsi 70 kugeza 90 | hafi 40 t/ha mu murima usanzwe (open field) | whitefly (udusimba duto tw'umweru tuguruka), thrips, bacterial wilt |

| Igihingwa | Intera yo gutera (cm) | Isarura rya mbere | Umusaruro | Ibyo kwitondera |
|---|---|---|---|---|
| Intoryi (eggplant) | 75 × 50 | iminsi 75 kugeza 110 | kg 10 kugeza 12 ku murongo wa m 10 [CRP-70] | fruit borer (inyo yinjira mu ntoryi), bacterial wilt |
| Dodo (amaranth) | 40 × 30 | iminsi 25 kugeza 30 nyuma yo kubiba, hanyuma buri cyumweru | kugeza kuri 10 t/ha z'amababi [CRP-71] | aphids (udusimba duto tunyunyuza ibimera) |
| Sukuma (kale) | 60 × 40 kugeza 60 | iminsi 42 nyuma yo kwimura ingemwe | hafi 15 t/acre [CRP-69] | diamondback moth, black rot |
| Imiteja (French beans) | 30 × 15 | iminsi 42 kugeza 56 nyuma yo kubiba | ubwoko bugufi (bush) 5 kugeza 6 t/ha | bean fly (isazi ntoya, inyo zayo zangiza uruti), rust (ingese ku mababi) |
| Concombre (cucumber) | 100 × 50 | iminsi 60 | kg 8 kugeza 10 ku murongo wa m 10 [CRP-70] | aphids, downy mildew |
| Epinari na bette (spinach and chard) | 45 kugeza 60 × 20 kugeza 25 | iminsi 60 kugeza 90 | ibimera 40 kugeza 50 ku murongo wa m 10 [CRP-70] | leaf miners (inyo nto zicukura imbere mu mababi) |
| Ibirungo (herbs) | persil (parsley) 40 × 15, coriandre (coriander) imirongo 15 | coriandre iminsi 28, persil 100 | uturima duto hafi y'igikoni gikuru | |

Intera, igihe, umusaruro, ibyonnyi n'indwara biva mu gitabo cy'imboga cya RAB [V CRP-68], keretse aho handitswe ukundi.

## Amazi

1. Mu mezi y'izuba ryinshi, buri m² y'akarima ikenera litiro 4.1 kugeza 5.3 ku munsi: 10 kugeza 13 m³ ku munsi ku murima wose [E CRP-75, LOC-54].
2. Ayo mazi akubye hafi inshuro icumi ayo ibisenge bishobora gufata, bityo borehole cyangwa umuyoboro rusange (mains) ni byo bitunga umurima; mu gihembwe C, tera gusa ibyo amazi yemejwe n'igerageza ashobora kuhira.
3. Uhire mu museke cyangwa nimugoroba, ntuzigere uhira saa sita z'amanywa; sasira uturima ibyatsi byumye (mulch) kugira ngo amazi agume mu butaka.
4. Kuhira ubuso buri munsi ya ha 1 ntibisaba uruhushya rw'Ikigo gishinzwe umutungo kamere w'amazi (Water Resources Board) [V LOC-08].

## Kurwanya ibyonnyi

1. Simburanya ibihingwa (rotation): ntuzigere utera umuryango umwe w'ibihingwa (inyanya, puwavuro n'intoryi; ishu na sukuma; imiteja n'ibindi bishyimbo) mu karima kamwe ibihembwe bibiri bikurikiranye.
2. Zenguruka uturima kabiri mu cyumweru; randura kandi utwike igihingwa cyose kiraba mu buryo butunguranye (bacterial wilt).
3. Manika imitego ikoresha impumuro (pheromone traps) ya Tuta absoluta mu turima tw'inyanya.
4. Tera umuti gusa uwo umujyanama w'ucuruza inyongeramusaruro (agro-dealer) akubwiye ku cyonnyi wabonye, kandi wandike buri muti uteye.

## Inyongeramusaruro no kugurisha

- **Abatanga ibikoresho (suppliers)** (ntibaravugishwa; kuba bari ku rutonde ntibivuga ko bafite ibicuruzwa mu bubiko): Holland Greentech ku mbuto n'ibikoresho byo kuhira hakoreshejwe ibitonyanga, Davis & Shirtliff kuri pompe, n'abacuruza inyongeramusaruro b'i Bugesera bari ku rutonde rwa RICA [V CRP-74].
- **Ifumbire** y'imboga ihabwa nkunganire (subsidy) binyuze kuri *774#, ukoresheje indangamuntu (ID) na UPI y'ubutaka; imbuto zo ntizihabwa nkunganire [V CRP-73].
- **Ibiciro i Bugesera** (igiciro ku murima / igiciro cyo kudandaza, RWF kuri kg, Nyakanga kugeza Ukwakira 2026): inyanya 350 / 700, igitunguru 475 / 750, ishu 164 / 350, karoti 550 / 1,000, puwavuro 850 / 1,500, imiteja 550 / 900, dodo 220 / 400 [V MKT-60]. I Kigali bagura ku giciro kiri hejuru: gurisha ibisagutse i Kigali cyangwa ku bashyitsi b'ikigo (compound).

## Ibigaragaza ko byarangiye

1. Ibisubizo by'igerageza ry'ubutaka n'iry'amazi byagarutse, kandi uturima twashyizwemo ishwagara n'ifumbire nk'uko ibyo bisubizo bibisaba.
2. Uturima mirongo ine twatewe, amatiyo y'ibitonyanga arakora, kandi buri kenge gatonyanga gatanga amazi.
3. Igikoni gikuru gihabwa urutonde rw'ibyasaruwe buri gitondo, kandi buri kilo gisaruwe mu murima kirandikwa.

## Ibibazo bitarasubizwa

1. Borehole itanga amazi angana iki kuva muri Kamena kugeza muri Kanama, kandi ayo mazi akwiriye gukoreshwa mu kuhira?
2. Ni nde uzakora mu murima buri munsi: umuhinzi umwe, cyangwa babiri?
3. Ni abahe baguzi b'i Kigali bazagura ibisagutse, kandi ku kihe giciro?